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Belagavi, Karnataka-590 018

A

Mini Project Report

On

AUDIO TRANSMISSION USING LI-FI

Submitted

in partial fulfillment of the requirement for the award of the Degree of

BACHELOR OF ENGINEERING

in

ELECTRONICS AND COMMUNICATION ENGINEERING

by

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DECLARATION

We the members of the mini project team, studying in the V semester of Electronics and Communication Engineering, Vidyavardhaka College of Engineering, hereby declare that the entire mini project titled **"AUDIO TRANSMISSION USING LI-FI""** has been carried out by us independently under the guidance of **Mr. Praveena KS,** Assistant Professor**,** Department of Electronics and Communication Engineering, Vidyavardhaka College of Engineering. This mini project work is submitted to the Visvesvaraya Technological University, Belagavi, in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Electronics and Communication Engineering** during the academic year 2023-2024.

This mini project report has not been submitted previously for the award of any other degree or diploma to any other Institution or University.

Place: Mysuru

Date: 07-03-2024

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ABSTRACT

Due to the radio wave problem with the medical instruments the “Wi-Fi” technology become a problem facing this challenges due to the interference of radio waves which may be harmful for the patient. Multiple accesses “Wi-Fi” is have many challenges starting from the capacity, availability, efficiency and security. Moreover WiFi transmit radio waves that may affect on medical instruments. In this paper a proposed transmitter and receiver based on “Li-Fi” technology communication by using “LEDs” which is line of sight only was done. A simple design to the “Li-Fi” is represented. The abstract of an audio transmission system using LiFi (Light Fidelity) involves leveraging visible light communication to transmit audio signals wirelessly. LiFi utilizes LED light sources to modulate data, enabling high-speed communication. This technology offers potential advantages, such as increased bandwidth, security, and reduced electromagnetic interference compared to traditional wireless communication methods. The abstract likely outlines the key components and benefits of this LiFi-based audio transmission system

CHAPTER-1

INTRODUCTION

Over the past few years, there has been a rapid growth in the utilization of the RF region of the electromagnetic spectrum. This is because of the huge growth in the number of mobile phones subscriptions in recent times. This has been causing a rapid reduction in the free spectrum for future devices. Hence, an alternative means to wireless communication is necessary to accommodate the exponentially increasing wireless traffic demand. Visible light communication systems provide an alternative to the current standards of wireless transfer of information, using light from LEDs as the communication medium.

1.1 BACKGROUND

The term Li-Fi was Invented by the German Professor at the University of Edinburgh, Harald Haas and it refers to light based communications technology that delivers a high-speed, bidirectional networked mobile communication which is similar to WiFi. Light-fidelity (Li-Fi) operates in the visible light spectrum of the electromagnetic spectrum i.e. it uses visible light as a medium of transmission rather than the traditional radio waves. Li-Fi is the transmission of data using visible light. This is done by sending data through an LED light bulb that varies in intensity at a speed much faster than the human eye can follow. If the LED is on, the photo detector registers a binary one; otherwise, it's a binary zero that is registered.

1.2 INTRODUCTION

Communication can be defined as the transmission of data, audio and signals from the source to distention or multi destination points, a huge number of technology develop to handle this process with high capacity, free of errors, reliability and serve multi users at the same time with respect to the overall performance.

"Li-Fi" based on the visible light communication technology and it is a system that is considered wireless communications with high speed. "Li-Fi" uses common uses "LED" for sensing data, at a high speed reach "224" Gbps.

"Li-Fi" and "Wi-Fi" are similar both transfer data electromagnetically but "Wi-Fi" uses radio waves in state of "Li-Fi" that uses visible light to send data. An "LED light" based on semiconductors that uses a direct power direct current supply. It can dimed up and down at very high speeds, and invisible for human eye.

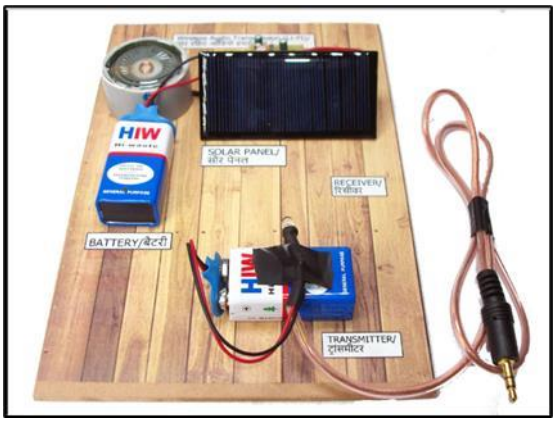


Figure.1.1: Audio transmission using Lifi

Figure.1 shows the structure of audio transmission using lifi and the transmission of signal and detection

1.3 PROBLEM DEFINITION/PROBLEM STATEMENT

The conventional methods of audio transmission, primarily reliant on radio frequency-based technologies, face challenges such as spectrum congestion, susceptibility to interference, and limited bandwidth. To address these issues and explore innovative solutions, there is a need for research and development in the field of LiFi (Light Fidelity) audio and data transmission. LiFi leverages visible light communication to offer a secure, high-speed, and interference-free alternative.

1.4 OBJECTIVE

To design and implement a robust Li-Fi communication system that utilizes LED lights for audio transmission.

1.5 EXISTING SYSTEM

WiFi and Bluetooth are the two consistent wide range sources used by multiple applications today. However, these methods use radio frequency spectrum and noise of the signal is very high. Other inconveniences of these methods are special equipment requirement, high power consumption, and high cost. Secure data transmission is not available here. WPS key encryption can be provided and hacking can also be done easily Since it uses radio wave transmission it is harmful to health

CHAPTER-2

METHODOLOGY

The principle of Li-Fi is based on sending data by amplitude modulation of the light source in a well-defined and standardized way. LEDs can be switched on and off faster than the human eyes can detect since the operating speed of LEDs is less than 1 microsecond. This invisible on-off activity enables data transmission using binary codes.

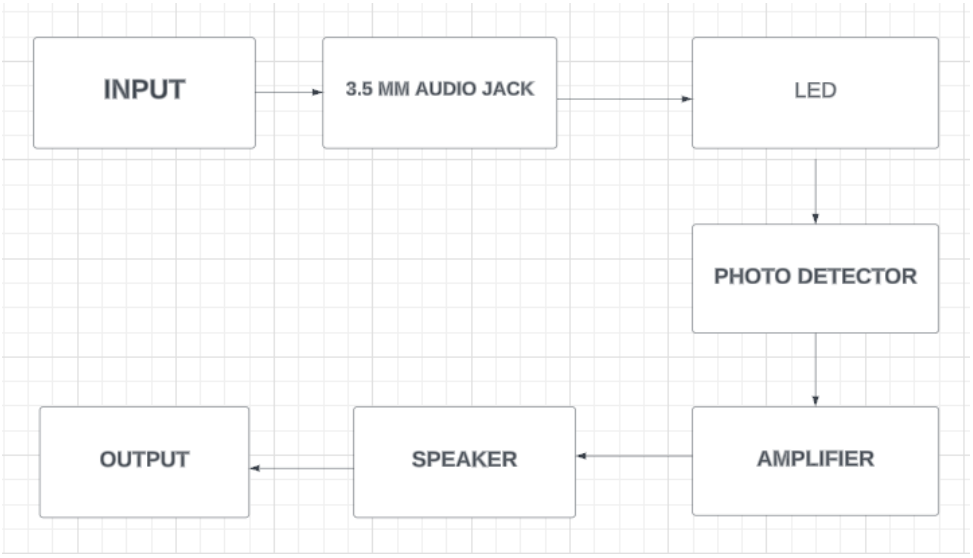


Figure.2.1: Block Diagram of Lifi technology

Figure.2.1 shows the block diagram of signal driving from input to output. In the process of an audio communication through the visible light, on the transmitter side an audio signal (any music from mobile) is used as the input signal. This signal is fed to the LED. The light signal from the LED varies according to the intensity of an audio signal. At the receiver side light dependent resister will receive the light signal and correspondingly generate an electrical signal proportional to it. This electrical signal is processed by a low voltage audio power amplifier, which is then fed to a speaker and it produces the audio signal which was at the input of the transmitter side.

INPUT: Steps for building the Transmitter:

1. Connect the Negative (-ve) Terminal of your LED with the GROUND terminal of the Headphone Jack

2. Now, Connect the Resistor to the Positive (+ve) Terminal of the LED

3. Connect the Positive (+ve) Terminal of your 9 Volt Battery with the Resistor

4. Now at last connect the Negative (-ve) Terminal of the Battery with the common wire of LEFT and RIGHT Terminals from the 3.5 mm Jack to Complete the Circuit.



Figure 2.2 : audio transmitter

OUTPUT: Steps for making the Receiver:

1. Solder the wires to the Positive (+ve) and Negative (-ve) Terminals of the Solar Panel

2. Now, Connect the Negative (-ve) Terminal to the GROUND Terminal of the other Jack

3. The Positive (+ve) Terminal is left so connect the Positive (+ve) Terminal of the Solar Panel to the Common wire of LEFT and RIGHT Terminals from the jack to complete the Circuit.

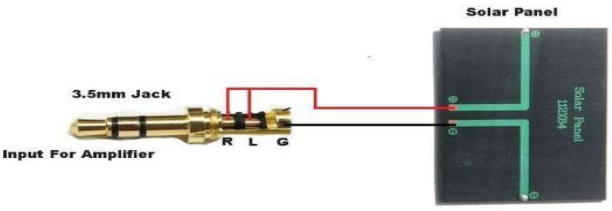


Figure 2.3 : Audio Reciver

2.1 HARDWARE AND SOFTWARE REQUIRMENTS

Software Requirements:

2.1.1 Audio playing application:

Audio playing software on mobile devices processes digital audio files into analog signals using a digital-to-analog converter (DAC). Once converted, the analog signals are either amplified and routed to the built-in speakers or transmitted through the headphone jack to connected earphones or external speakers. For wireless transmission, Bluetooth technology is employed to send the audio data to compatible devices. This seamless process ensures users can enjoy their favorite music or audio content with clarity whether through wired connections or wireless technology.



Figure 2.1.1 Audio playing software in mobile

Audio will be played in mp3 format in the audio playing software.Many apps or available to play the mp3 audio files or from local storage of mobile

Hardware Requirements:

2.1.2 LED(Light emitting diode):

LED (Light-Emitting Diode) works by converting electrical energy into light through electroluminescence. In LiFi projects, LEDs serve as transmitters, emitting modulated light signals carrying data. For transferring audio files from mobile devices in LiFi, the audio data is converted into digital signals and modulated onto the LED's light. The LED rapidly switches on and off, encoding the digital information. On the receiving end, a photodetector, like a 6V solar panel, captures the modulated light signals and converts them back into digital data. Transfer speeds for basic setups using a 1.5V LED and a 6V solar panel can range from a few Kbps to tens of Kbps. Data is transferred in digital format, ensuring accuracy and fidelity. The speed depends on factors like modulation techniques, hardware

capabilities, and environmental conditions. Advancements in technology may lead to higher transfer rates as LiFi systems evolve.



Figure 2.1.2 LED

2.1.3 Resistors:

A 220-ohm resistor is commonly used in conjunction with a 9V battery and a 1.5V LED to limit the current flowing through the LED and protect it from damage due to excessive current. LEDs typically have a forward voltage drop around 1.5V. When connected directly to a 9V battery without a resistor, the LED may draw too much current, potentially causing overheating and failure. The 220-ohm resistor acts as a current limiter, ensuring that the LED operates within its safe operating range. By Ohm's Law (V=IR), with a 9V supply and a 220-ohm resistor, the current flowing through the LED is approximately (9V - 1.5V) / 220 ohms = 0.036A or 36mA. This setup provides proper illumination for the LED while protecting it from excessive current flow.



Figure 2.1.3 A 220ohm Resistor

2.1.4 Connecting wires:

Wires are used to make connection between components which forms a conducting path.

2.1.5 Power source:

A 9V battery is a compact power source commonly used in electronics projects. It provides a stable output voltage of approximately 9 volts, making it suitable for powering small-scale circuits and devices. Its portability and consistent voltage output make it ideal for applications where mains power is unavailable or impractical. Engineers and hobbyists often rely on 9V batteries to power prototypes, sensors, and educational demonstrations.



Figure 2.1.4 A 9V General purpose Battery

2.1.6 Solar panel:

Solar panels work by converting sunlight into electricity through the photovoltaic effect. When photons from sunlight strike the solar cells, they knock electrons loose, creating a flow of electricity. This electricity is then captured and converted into usable power by the solar panel's internal components. Solar panels provide clean and renewable energy for various applications, including residential and commercial electricity generation.



Figure 2.1.5 A Solar panel-6v

In a LiFi audio transmission project, a 6V solar panel receives audio signals from a 1.5V LED through modulation of light intensity. The LED emits light modulated with audio data, which is captured by the solar panel's photodetector. The varying intensity of light induces a fluctuating voltage output in the solar panel corresponding to the audio signal. This voltage signal is then amplified and processed to extract the audio data, which can be directly fed into a speaker or further decoded for playback. Thus, light intensity variations from the LED effectively encode audio data for transmission to the solar panel, where it is converted into electrical signals for audio playback.

2.1.7 Speaker:

In a LiFi audio transmission system, the electrical signals generated by the solar panel in response to modulated light from the LED are typically weak. Therefore, to amplify these signals for audio playback, an amplifier circuit is employed. This amplifier circuit takes the weak electrical signals from the solar panel as input and boosts their strength to levels suitable for driving a speaker.

The amplified signals are then sent to the speaker, which converts the electrical signals back into sound waves. The speaker's cone vibrates in response to the varying electrical signals, producing audible sound waves corresponding to the original audio data encoded in the light signals.

Overall, the amplifier plays a crucial role in LiFi audio transmission systems by amplifying the weak electrical signals generated by the solar panel, allowing for clear and audible playback through the speaker.



Figure 2.1.6 Speaker with inbuilt amplifier

In a LiFi project where audio signals are transmitted from a solar panel to a 10W mini speaker via an aux input, the varying voltage audio signal generated by the solar panel needs to be amplified to drive the speaker effectively. Let's consider hypothetical values for demonstration:

1. Solar Panel Output: The solar panel generates a varying voltage audio signal in response to modulated light from the LED. Let's say the peak-to-peak voltage of this signal ranges from 0.1V to 0.5V.

2. Amplification: The audio signal from the solar panel is weak and needs amplification. A 10W mini speaker typically requires a certain input voltage to produce the desired output sound level. the amplifier provides a gain of 20x.

3. Amplified Signal: With a gain of 20x, the varying voltage audio signal from the solar panel is amplified to a range of 2V to 10V peak-to-peak.

4. Speaker Output: The amplified signal is then fed into the mini speaker's aux input. The speaker's internal circuitry converts the electrical signal back into sound waves, producing audio with sufficient volume and clarity.

Overall, the amplifier increases the voltage level of the audio signal from the solar panel to drive the 10W mini speaker effectively, ensuring audible playback of the transmitted audio data in the LiFi project.

Amplifier used in speaker

Mini speakers typically use Class-D amplifiers due to their efficiency and compact size. Class-D amplifiers are known for their high efficiency, making them ideal for small devices like mini speakers where power consumption and space are critical factors. These amplifiers use pulse-width modulation (PWM) to convert the audio signal into a series of pulses, which are then amplified and filtered to reconstruct the original audio signal. This allows for efficient power usage and minimal heat generation, which is important in small form factor devices.

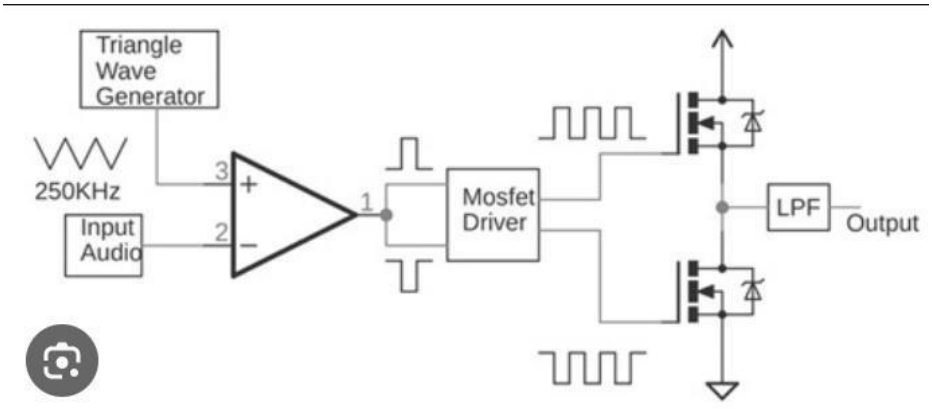


Figure 2.1.7 Class d amplifer circuit

A Class-D amplifier efficiently amplifies audio signals by rapidly switching output transistors on and off. It converts the analog audio signal into a series of pulses using Pulse Width Modulation (PWM), switches these pulses to control the output transistors, filters the signal to remove high-frequency components, and then sends the amplified audio signal to the speaker. This design results in high efficiency, low heat generation, and compact size, making Class-D amplifiers popular for portable devices and audio systems.

CHAPTER-3

APPLICATIONS

There is a wide necessity for data transfer and by the end of the day every field involves the use of technologies. One such technology is Li-Fi which can have its applications extended in areas where the Wifi technology lack its presence like medical technology, powerplants and various other areas where Li-Fi proved it excellence of the undersea awesomeness.

At present its applications ate beyond imagination but still if to think about few then they are:

- Can be used in the places where it is difficult to lay the optical fiber like hospitals. In operation theatre LiFi can be used for modern medical instruments.
- In traffic signals LiFi can be used which will communicate with the
- LED lights of the cats and accident numbers can be decreased.
- Thousand and millions of street lamps can be transferred to LiFi lamps to transfer data.
- In aircraft LiFi can be used for data transmission.
- It can be used in petroleum or chemical plants where other transmission or frequencies could be hazardous.
- Li-Fi can also be used in underwater systems for audio communications and device control.
- Li-Fi audio transmission can be used in petrochemical industries automations where use of radio spectrum is very dangerous.

CHAPTER-4

ADVANTAGES AND LIMITATIONS

Advantages:

- Faster Data Transmission than Wi-Fi.
- Easy and Inexpensive to Deploy.
- Security Due to the Limitations of Light.
- Immune from Electromagnetic interferences.
- Low Cost
- Portability
- Low bit error rate
- High efficiency
- Consumes less energy

Limitation:

- The Lifi limitations that unable to pass through obstacles (such as walls). Moreover the Lifi requires the light is on all the time for connectivity, that is mean the LEDs light is required to be on at 24 h a day.

CHAPTER-5

RESULTS AND DISCUSSION

Results:

- Successful establishment of LiFi audio transmission between LED transmitter and solar panel receiver.
- LED emitted modulated light signals carrying audio data.
- Solar panel received and converted light signals into electrical signals.
- Amplified and processed signals drove a speaker, resulting in audible playback of transmitted audio.

Discussion:

- Demonstrated feasibility of light-based audio transmission with potential advantages over RFmethods.
- Short transmission range observed; signal quality degraded with distance.
- Challenges include mitigating effects of ambient light and environmental factors.
- Future improvements could focus on enhancing LED intensity, photodetector sensitivity, andsignal processing techniques for better performance and reliability.

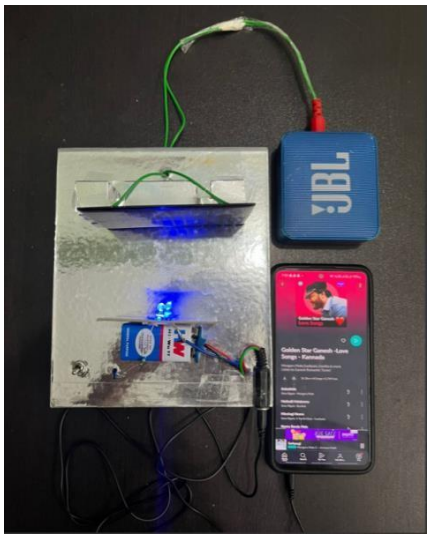


Figure 5.1 implemented model

CONCLUSION

The possibilities are numerous and can be explored further. If his technology can be put into practical use. every bulb can be used something like a Wi-Fi hotspot to transmit wireless data and we will proceed toward the cleaner. greener. safer and brighter future. The concept of Li-Fi is currently attracting a great deal of interest, not least because it may offer a genuine and very efficient alternative to radio-based wireless. As a growing number of people and their many devices access wireless internet, the airwaves are becoming increasingly clogged, making it more and more difficult to get a reliable. high-speed signal. This may solve issues such as the shortage of radio-frequency bandwidth and also allow internet where traditional radio based wireless isn't allowed such as aircraft or hospitals. The main shortcoming however is that it only work in direct line of sight.

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